(4) $-\sin x - 2\sec^2 x$ (5) $3t^2\cos t - t^3\sin t$ (6) $4\sec t\tan t + \sec^2 t$

(7) $e^x(\cos x + \sin x)$ (8) $\dfrac{x\sec^2 x - \tan x}{x^2}$ (9) $\dfrac{1}{1+\cos x}$

(10) $\dfrac{(1-x)\cos x + (1+x)\sin x}{(\sin x + \cos x)^2}$ (11) $\cos x + \sin x$ (12) $\dfrac{x\cos x - 2\sin x}{x^3}$

(13) $\tan\theta\sec\theta + \cos\theta + \sin\theta$ (14) $-\dfrac{(1+\cos^2 x)}{\sin^3 x}$ (15) $x\cos 2x + \sin x\cos x$

(16) $e^{-x}\left[\dfrac{1}{x} - \log x\right]$ (17) $e^{-3x}\left[-3(x^2+5)\log(1+x) + \dfrac{x^2+5}{1+x} + 2x\log(1+x)\right]$

(18) $\dfrac{\pi}{180}\cos\dfrac{\pi}{180}x$ (19) $\dfrac{\log_{10} e}{x}$

பயிற்சி 10.3

(1) $5(2x+4)(x^2+4x+6)^4$ (2) $3\sec^2 3x$ (3) $-\sec^2 x\sin(\tan x)$

(4) $x^2(1+x^3)^{-\frac{2}{3}}$ (5) $\dfrac{1}{2\sqrt{x}}e^{\sqrt{x}}$ (6) $e^x\cos(e^x)$

(7) $7(3x^2+4)(x^3+4x)^6$ (8) $\dfrac{3}{2}\left(t-\dfrac{1}{t}\right)^{\frac{1}{2}}\left(1+\dfrac{1}{t^2}\right)$ (9) $\dfrac{1}{3}\sec^2 t(1+\tan t)^{-\frac{2}{3}}$

(10) $-3x^2\sin(a^3+x^3)$ (11) $-my$ (12) $20\sec 5x\tan 5x$

(13) $\dfrac{8(2x-5)^3}{(8x^2-5)^4}[-4x^2+30x-5]$ (14) $\dfrac{8x^3+14x}{3(x^2+2)^{\frac{2}{3}}}$ (15) $e^{-x^2}[1-2x^2]$

(16) $-\dfrac{3t^2}{2(t^3+1)^{\frac{3}{4}}(t^3-1)^{\frac{5}{4}}}$ (17) $\dfrac{14-3x}{2(7-3x)\sqrt{7-3x}}$ (18) $-\sin x\sec^2(\cos x)$

(19) $\sin x(1+\sec^2 x)$ (20) $\dfrac{5^{-\frac{1}{x}}(\log 5)}{x^2}$ (21) $\dfrac{\sec^2 x}{\sqrt{1+2\tan x}}$

(22) $3\sin x\cos x(\sin x - \cos x)$ (23) $-k\sin kx\sin(2\cos kx)$ (24) $-6\sin 2x(1+\cos^2 x)^5$

(25) $\dfrac{3e^{3x}+2e^{4x}}{(1+e^x)^2}$ (26) $\dfrac{2\sqrt{x}+1}{4\sqrt{x}\sqrt{x+\sqrt{x}}}$ (27) $e^{x\cos x}[\cos x - x\sin x]$

(28) $\dfrac{4\sqrt{x}\sqrt{x+\sqrt{x}}+2\sqrt{x}+1}{8\sqrt{x}\sqrt{x+\sqrt{x}}\sqrt{x+\sqrt{x+\sqrt{x}}}}$ (29) $\dfrac{\cos\left(\tan\sqrt{\sin x}\right)\sec^2\left(\sqrt{\sin x}\right)\cos x}{2\sqrt{\sin x}}$ (30) $\dfrac{-2}{1+x^2}$

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